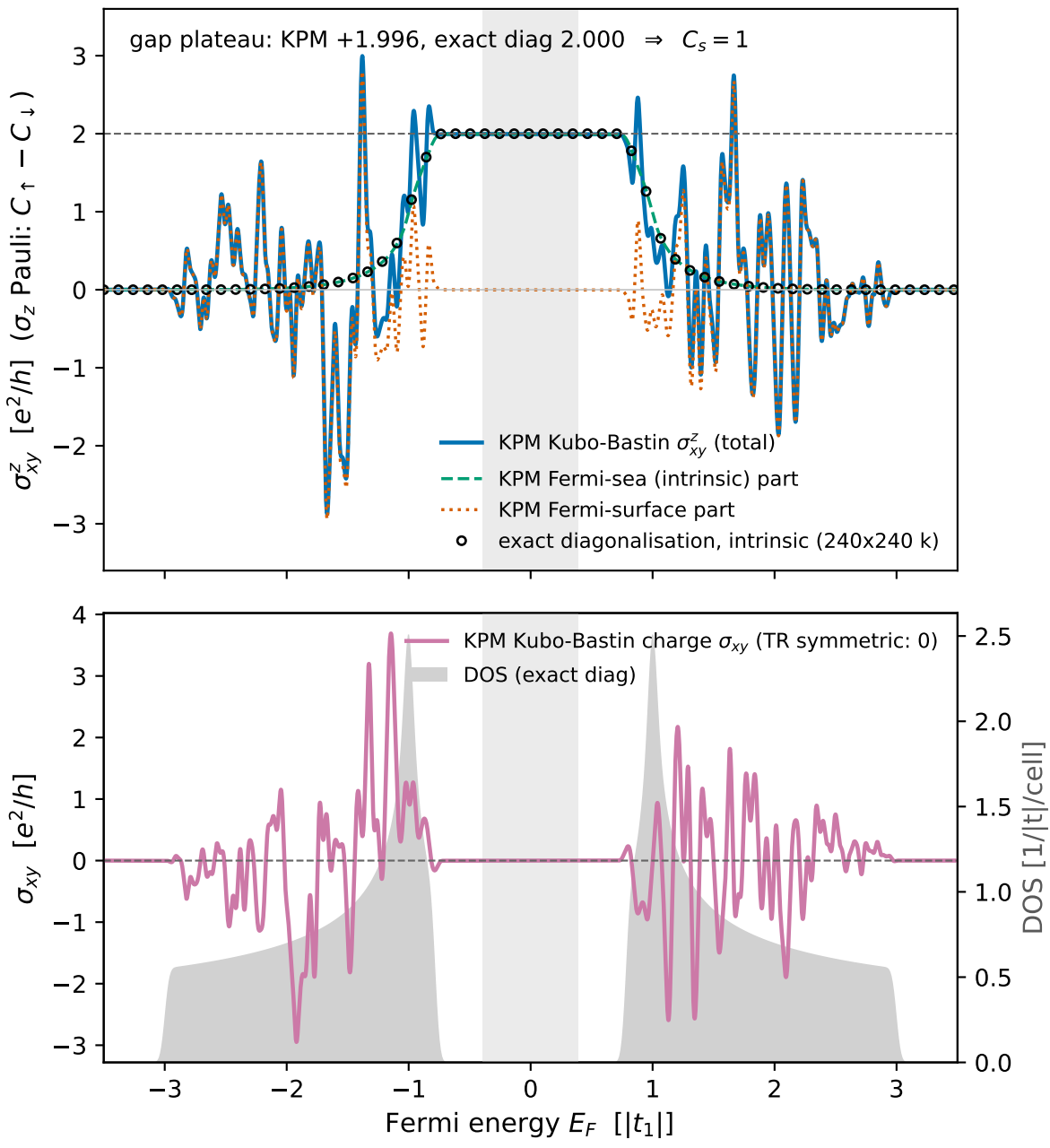

Quantum spin Hall effect, Kane-Mele model (linear-scaling KPM, lsquant)
$t_1 = -1$, $\lambda_{SO} = 0.15$, 60x60 cells, M=512, 16 random vectors, 30 meV
gap plateau: KPM +1.996, exact diag 2.000 ⇒ $C_s = 1$
σ^z_{xy} [e²/h] (σ_z Pauli: $C_\uparrow - C_\downarrow$)
KPM Kubo-Bastin σ^z_{xy} (total)
KPM Fermi-sea (intrinsic) part
KPM Fermi-surface part
exact diagonalisation, intrinsic (240x240 k)
KPM Kubo-Bastin charge σ_{xy} (TR symmetric: 0)
DOS (exact diag)
σ_{xy} [e²/h]
DOS [1/|t|/cell]
Fermi energy E_F [$|t_1|$]